

Research paper

Credit Risk Drivers in Hotel Sector – Hotels' Location and Distance-to-Default

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ABSTRACT

This paper examines the determinants of credit risk in the hotel sector and discuss the importance of location as a distance-to-default driver. Using public data from multiple sources and developing different innovative types of location indicators based on geospatial information, it analyzes hotels between 2010 and 2015. Credit risk is assessed by using a well-established Z-score based on financial indicators and its relationship with geospatial distances to touristic places. The research highlights that location plays a significant role in assessing credit risk. Hotels located closer to major tourist places such as beaches, castles, and churches generally exhibit lower risk profiles. However, the relationship is not linear and sometimes even inverted, each segment showing different shock sensitivities. The results also highlights the importance of tourist attraction critical mass concept for hotel financial performance and viability and demonstrates that distress signals are location-contingent. These findings provide useful insights for financial institutions, investors, and policymakers seeking to understand risk exposure in the hotel sector.

1. Introduction

The hotel sector is a fundamental pillar of the global travel and tourism industry, significantly contributing to economic growth, employment, and international trade. In 2023, destinations around the world earned 1.5 trillion dollars in international tourism receipts ([World Tourism Organization, 2024](#)). Tourism directly accounted for 3% of global GDP, amounting to 3.4 trillion dollars. In the European context, the region remains the world's largest tourism destination. In 2023, Europe welcomed approximately 707 million international arrivals ([World Tourism Organization, 2024](#)). This notable performance underscores the hotel sector's critical importance to the region's economic and social development.

Enterprises in the hotel sector require substantial investments to sustain growth and development. Consequently, credit risk directly affects a variety of stakeholders, including investors, financial institutions, and policymakers. Banks must carefully evaluate historical data to mitigate the risks of non-performing loans, which arise when enterprises fail to meet repayment obligations. In the context of the hotel sector, this involves considering a combination of factors, including its sensitivity to economic cycles and fluctuating demand patterns, substantial investments in fixed assets, and how low occupancy rates may compromise service quality. Moreover, hotels are particularly vulnerable to external shocks—such as the 2008-2009 Global Financial Crisis and the 2020-2021 COVID-19 pandemic crisis—which raises uncertainty and volatility in client behavior. Another critical dimension is seasonality, which increases economic instability by causing strong fluctuations in demand across the year. This leads to lower capacity use and higher fixed costs during off-peak periods, while peak seasons are often linked to higher prices and temporary employment. As noted by [Cuccia and Rizzo \(2011\)](#), seasonality generates significant private and social costs that are usually greater than its few benefits, affecting private producers (i.e. hotels, restaurants), final consumers of the destination (tourists and residents), and workers across the tourism sector. A second critical dimension is location, as hotels situated in central areas—such as city centers, coastal regions, or near major tourist attractions like castles, churches, or beaches—tend to benefit from higher and more stable occupancy rates. [Alvarez Leon et al. \(2021\)](#) support this by showing that proximity to urban and coastal attractions enhances hotel performance. A third important aspect is market competition, which also influences hotel outcomes. According to [Lado-Sestayo, Vivel-Búa, and Otero-González \(2017\)](#) factors such as geographical location, local competition, and agglomeration economies significantly affect hotel performance. In particular, the geographic location of a hotel is a significant driver of credit risk, it directly affects its exposure to demand, accessibility, and proximity to tourist, business, or cultural structures. Therefore, locational characteristics should be incorporated into credit risk assessments, as they directly influence a hotel's ability to generate stable revenues and maintain long-term financial solvency. By balancing the financial needs of the hotel sector with effective risk management strategies, potential threats can be proactively identified and assessed, thereby enhancing decision-making and ensuring sustainable outcomes.

The objective of this research is to analyze in detail the importance of geographical location and the different types and characteristics of touristic attractions as risk drivers associated with hotels performance and credit risk. While hotel location and operational characteristics independently influence financial performance (Chung & Kalnins, 2001), emerging evidence suggests these effects are not additive but interactive (Rigall-I-Torrent et al., 2011; Soler & Gemar, 2018). Recent studies document that hotel size (Xu and Xiao, 2020), quality positioning (Rigall-I-Torrent et al., 2011; Assaf et al., 2015), and operational efficiency metrics (Yang et al., 2014; Yang et al., 2015) demonstrate location-contingent effects. Yet, credit risk research has largely neglected these conditional relationships, treating hotel characteristics as universally predictive despite accumulating evidence of context-dependent effects. We consider variables related to hotels' location, types of touristic attractions, hotels' financial indicators, market competition and hotels' characteristics.

The sample consists of 1,063 hotels (444 micro, small, and medium-sized and 619 large hotels) representing almost the entire universe of hotels in Portugal and includes information on 882 tourist points. According to the RNET database from Turismo de Portugal, MSMEs account for 41% of registered hotels in Portugal, while BMEs represent 59%. Portugal offers several unique characteristics that make it an excellent natural laboratory for studying how hotel location influences credit risk. It provides uniform macroeconomic conditions (same monetary policy, fiscal framework, labor market) and contains remarkable geographic diversity within short distances. The country offers a varied range of tourism types, including cultural tourism, religious tourism, and beach tourism across the entire continental territory and islands. This allows to isolate location effects while controlling broader economic variables. The stark contrasts between thriving urban and coastal areas and struggling interior regions create natural experiments where hotels face dramatically different demand conditions based purely on location, not national economic factors. Tourism intensity varies also enormously by region, where some areas are heavily international tourism-dependent while other areas rely more on domestic travelers or struggle with low demand. International tourism is more vulnerable to external shocks creating different demand volatility patterns by location.

This study covers the period 2010–2015, a period that follows the 2008 Global Financial Crisis and integrates the impact of the 2010-2012 Euro-Area Sovereign Crisis in the hotel sector. This period offers exceptional research value for several compelling reasons. Hotels faced a "double hit" scenario—recovering from one crisis while entering another—creating ideal conditions to study how external shocks compound and affect credit risk in vulnerable sectors. In addition, Portugal was one of the hardest-hit countries, requiring an EU-IMF bailout in 2011. The country faced severe austerity measures, high unemployment, and credit market disruptions. For hotels, this meant domestic demand collapsed while international perception of Portugal as a destination also suffered, creating extreme stress conditions perfect for studying credit risk, default and survival patterns. If location factors matter for hotel credit risk during this period of extreme stress, they likely matter under most conditions. Portuguese banks faced severe difficulties during this period, with credit markets freezing (Iyer et al, 2014) and lending standards tightening dramatically. This allows researchers to examine how location affects not just hotel performance but also access to credit and refinancing ability during financial system stress—critical for understanding credit risk. This stress-test environment provides more robust and generalizable findings. The crisis period generated substantial hotel defaults, bankruptcies, and debt restructurings—providing the credit events necessary for robust statistical analysis of default determinants. Studying "normal" periods with few defaults' limits research insights. Moreover, the period largely predates the massive short-term rental disruption that hit Portugal after 2015, allowing cleaner analysis of traditional location and economic factors without the confounding effect of platform competition and 2015-2019 rapid tourism growth.

This study contributes to the existing literature in two main folds. First, this study contributes to the emerging research debate on the intersection of spatial economics, hospitality performance and credit risk modeling. Few studies integrate location's impact on revenue and occupancy into credit risk studies to show how geographic factors directly influence credit quality scores. Foundational work has been established primarily by Vivel-Búa, Lado-Sestayo and Otero-González, however empirical evidence linking specific market segment exposures (leisure beach, cultural tourism, and religious tourism) to credit scoring is scarce, particularly for the hospitality sector. This study provides unique detailed insights into the location of hotels and their proximity to different types and quantity of tourist attractions such as beaches, churches and castles influence financial performance and credit risk. This unique granularity could reveal non-linear relationships (e.g., optimal distance thresholds) missed by coarser geographic categorizations. Location is a structural, time-invariant characteristic that may create underlying vulnerability or resilience, therefore by using different types and quantity of tourist attractions demonstrates "where" and "how" those tourist attractions would be significant. In the same context, this study presents methodological advancements by integrating spatial data (GIS coordinates, distance calculations) into financial risk modelling. Previous studies (Vivel-Búa et al., 2018) rely on the concept of city center's location, however this study goes further by using public data to calculate exact distances from hotels to georeferenced tourist points, providing a rigorous methodological definition. This represents a methodological innovation and allows a more precise assessment of location effects, demonstrating how spatial analytical techniques can enhance credit risk assessments in the hotel sector. The hospitality literature has long recognized beach proximity as a critical location attribute for coastal hotels (Espinete et al., 2003; Lei & Lam, 2015). Traditional hedonic pricing studies document significant beach proximity premiums (Schamel, 2012), while more recent work identifies non-monotonic relationships due to trade-offs between accessibility benefits and competition costs (Rigall-I-Torrent & Fluvía, 2011). However, no prior study has examined how beach distance patterns affect hotel credit risk, nor distinguish between immediate competitive proximity and regional beach

infrastructure density. We hypothesize that immediate beach proximity captures competitive crowding effects, while regional beach density reflects broader tourism ecosystem quality. Second, this study contributes to the credit literature debate whether structural factors like location matter more: during crises (when financial cushions erode and fundamental economics dominate), or during normal times. Evidence on whether and how location protections or vulnerabilities manifest during severe economic crises is limited. The 2010–2015 period offers rare opportunity to observe location effects under extreme stress—when credit risk crystallizes most dramatically.

This study focuses on a crisis-period and directly addresses this debate, potentially showing that location effects amplify during stress. Moreover, the literature acknowledges hotels face demand volatility but rarely explores how location determines exposure to different volatility sources such as international vs. domestic shocks. While the corporate finance literature extensively documents ratio management in financial distress (DeAngelo et al., 1994; Asquith et al., 1994; Roychowdhury, 2006), and hospitality research has examined bankruptcy prediction using financial ratios (Gu, 2002; Kim & Gu, 2006), to our knowledge no prior study has explicitly examined the potential coordinated management of profitability margins and liquidity ratios in the hotel sector during financial stress. Moreover, the role of location viability in moderating these relationships remains unexplored. We address these gaps by document their coordinated management in the hotel sector and linking this behavior to location viability. In summary, this study contributes to both the study of the intersection of spatial economics, hospitality performance and credit risk modeling using granular spatial data in innovative ways, as well as the study of a crisis-period providing evidence on structural (location) versus cyclical (financial) risk determinants. The findings of this study aim to inform decision-making by investors, financial institutions, and policymakers, thereby enhancing strategic planning and promoting more sustainable financial and operational outcomes in the hotel sector.

The study is organized as follows: Section 2 presents a review of related literature. Section 3 outlines the data and methodology. Section 4 explains the results and discussion, and Section 5 concludes the study and suggests opportunities for future research.

2. Literature Review

The studies on the intersection of spatial economics (hotel location), hospitality performance, and credit risk modeling are scarce. This strand of the literature is addressed by some empirical studies demonstrating that proximity to heritage sites, train stations, airports, and coastlines increases revenues, mitigating financial risk. Lado-Sestayo, Otero-Gonzalez, Vivel-Búa, and Martorell-Cunill (2016) demonstrate that location is a key determinant of profitability in the Spanish hotel sector, with hotels situated in coastal areas and on islands achieving higher financial performance, largely due to the significance of sun-and-beach tourism in Spain. In a related study, Lado-Sestayo, Vivel-Búa, and Otero-González (2017), using a large dataset of Spanish hotels, find that after hotel attributes such as location, size, star rating, management (efficiency of assets, and efficiency of labor), explain the total revenue per available room. Their analysis highlights that geographical location, local competition, and agglomeration economies show significant effects on hotel outcomes. Extending this research, Lado-Sestayo, Vivel-Búa, and Otero-González (2020) confirm that location is connected to profitability drivers. J.Sánchez-Sánchez (2025) demonstrates that hotels located in Mediterranean destinations, the Balearic Islands, and the Canary Islands achieve superior profitability due to the attractiveness and accessibility of these locations, which lower exposure to seasonality and increase the consistency of cash flows. Moreover, locational effects are not limited to proximity to natural or cultural attractions but also include the influence of nearby areas. The research on Spanish hotels indicates that revenues are influenced not only by a hotel's own location but also by the characteristics of surrounding establishments. Specifically, hotels situated within proximity to points of tourist interests such as heritage sites, airports, and coastlines tend to achieve better economic outcomes. This suggests that location-driven advantages are amplified through neighborhood effects, as hotels located in areas with high tourist demand attract more tourists, stimulate complementary services, and collectively reinforce demand stability.

These findings align with the broader literature on spatial economics, which argues that hotels benefit from geographic location. Importantly, location also determines the ability of hotels to withstand external shocks, such as economic downturns or major events. For example, centrally located hotels or those near key attractions are better positioned to capture extraordinary demand surges during events and to cushion losses during periods of reduced tourism activity, thereby lowering their financial vulnerability. Collectively, the European evidence indicates that favorable location enhances occupancy rates, increases average revenue metrics, and stabilizes operating income, all of which contribute to stronger financial performance and lower credit risk. From a risk management perspective, this implies that locational characteristics should not be considered merely operational variables but strategic determinants of financial stability and creditworthiness. To systematically capture this spatial proximity, this study adopts a geographic information system (GIS)-based approach. Following the methodology applied by Sholeh et al. (2017), who used latitude and longitude data and the Google Maps API to locate and map business entities, this approach enables our research to calculate precise Euclidean distances (María et al., 2020) between hotels and key tourist attractions. This process ensures greater accuracy and consistency when incorporating geographic variables into credit risk models. In addition, this study draws also on agglomeration economies (Chung & Kalnins, 2001; Canina et al., 2005), which predicts inverted U-shaped relationships between spatial clustering and firm performance. While prior work focuses on general hotel clustering, we extend this logic specifically to coastal locations, where beach resources create natural agglomeration.

Moreover, this research considers traditional determinants already described in literature as controls, such as size, liquidity, leverage, and profitability. One key factor influencing credit risk in the hotel sector is the asset size of the establishments. In principle, larger hotels possess greater capacity to absorb and adapt to economic fluctuations affecting their operations (Sullivan, 1978) and tend to leverage their market position to gain competitive advantages (Ang et al., 1985). Operational efficiency likewise affects risk levels. Efficient resource utilization enhances the firm's capacity to generate revenues and profits, thereby mitigating credit risk. This effect has been corroborated in studies examining systematic risk within the sector (Kim et al., 2002, 2007, 2012). Profitability is also a relevant factor, as firms with higher returns on assets tend to have a reduced probability of financial distress and bankruptcy (M.-H. Chen, 2013). Metaxas and Romanopoulos (2023) provides a comprehensive review of 29 studies on hotel credit risk and financial distress, finding that debt and liability measures increase default risk while profitability and size reduce it. However, financial metrics can be misleading without operational context. Jensen (1986) showed that high profitability combined with excess cash retention signals agency conflicts and financial distress anticipation rather than strength. Managers anticipating distress engage in defensive financial management by simultaneously cutting costs to boost margins while hoarding liquidity for eventual restructuring needs. Opler et al. (1999), in the context of the precautionary cash motives, showed that firms in uncertain environments hold excess cash, and competition intensity is a key uncertainty driver, therefore cash hoarding signals financial vulnerability.

3. Data and Methodology

This study considers information at hotel level such as distances from premises to nearby tourist points, financial indicators, market competition and property and operational characteristics as key determinants. The main hypothesis is that hotels closer to major touristic attractions tend to have lower credit risk. There are some arguments that may support this hypothesis. Hotels near major tourist attractions typically benefit from consistent visitor flows, which can translate to more predictable occupancy rates and revenue streams, reducing volatility in cash flows and improving profitability and debt service capacity, key components of credit risk. Major tourist attractions may also allow a diverse customer base, for instance, leisure and business travelers (conferences, events), providing demand diversification that buffers against downturns in any single market segment. However, there are also important counterarguments to consider. Many tourist attractions have pronounced seasonal patterns and a hotel heavily dependent, for instance, on summer beach tourism might face significant cash flow volatility, potentially increasing credit risk despite proximity to attractions. Market saturation is also possible in areas near major attractions (for instance, hotels clustered around theme parks or major monuments such as churches or castles) with high hotel supply and intense competition, which can compress margins and occupancy rates. In addition, tourism-dependent hotels can be highly vulnerable to economic recessions (e.g. 2008-2009 Global Financial Crisis; 2010-2012 Euro-Area Sovereign Debt Crisis), travel disruptions (pandemics, terrorism concerns), or changes in travel patterns. Moreover, the popularity of tourist attractions can change over time due to trends, deterioration, changes in regulation, or new competing attractions emerging elsewhere.

A comprehensive approach to understanding the factors shaping hotel creditworthiness is developed by addressing the main research question: how do hotel location and nearby tourist attractions influence credit risk? In order to test empirically that hotels closer to major touristic attractions tend to have lower credit risk, this study controls for other confounding variables like financial indicators (size; profitability; liquidity; debt), market characteristics (e.g. market competition), property characteristics (e.g. hotel quality), management quality (e.g. brand strength; operator experience via years in business), geographic (NUTS II and III) and time controls. This study measures proximity quantitatively (distance in kilometers) and compares hotels' individual Z-Scores (as a proxy for credit risk) across different distance bands while controlling for those confounding factors. The relationship may not be linear. For instance, there may be an optimal distance zone where hotels capture tourist demand without facing extreme competition or the operational challenges of being in the most congested areas. The effect of hotel's proximity to tourist attractions from other factors that influence both hotel's location choice and credit risk is isolated, with a well-specified model that includes well-established controls to avoid omitted variable bias. Endogeneity concerns, i.e. hotels that may strategically locate near touristic attractions and touristic attractions that may develop in areas with hotel infrastructure, were also considered by using instrumental variables.

We use comprehensive information about the hotels registered in Portugal. Different data sources were used, namely: Turismo de Portugal, Visit Portugal website; Orbis BvD; INE, Statista, and Pordata. The Turismo de Portugal dataset provided extensive details about the characteristics of hotels. Orbis database provided financial information. We aimed to integrate hotel characteristics with their corresponding financial data. The integration of the datasets was made possible with common variables. For instance, the Tax Identification Number and geographic classification (EU Nomenclature of Territorial Units for Statistics – NUTS). The financial data from Orbis was matched with the corresponding hotels using their respective Tax Identification Numbers, ensuring accurate and cohesive dataset integration for the analysis. To ensure consistency between the datasets, we calculated the percentage of hotels whose Tax Identification Numbers matched between the Turismo de Portugal dataset and the Orbis dataset, focusing on the EU Nomenclature of Economic Activities (NACE) codes 55 and classified as legal entities. The analysis revealed a 95.33% compatibility rate, indicating that most hotels in the Turismo de Portugal dataset had corresponding financial data available in the Orbis dataset. This high degree of alignment allowed

us to proceed with our analysis without significant loss of data due to unmatched identifiers or missing financial information for the respective hotels. The NUTS II variable (the second level of the EU's NUTS hierarchy, a standard formal way to break down the EU's territory for statistical purposes) facilitated the linkage between the datasets containing information on Portuguese hotels and tourist points.

Sample Representativeness

Data from Turismo de Portugal was used to include all hotels from Portugal that opened in or before 2015, allowing the inclusion of 1,149 hotels. The focus of this study was specifically on analyzing hotels registered as firms, i.e. “legal entities” and excluding “individuals”. After filtering, our sample of Portuguese hotels was reduced to 1,063 (i.e. only less 86 hotels). Then, we categorized those hotels based on their size using the number of rooms (Devesa & Peñalver, 2013) as follows: micro-hotels in case of 20 or fewer rooms; small hotels for those between 21 to 50 rooms; medium hotels for those between 51 and 80 rooms; large hotels for those between 81 to 120 rooms; and macro-hotels for those with more than 121 rooms. Within this sample, filtered by NACE code 55, we identified 14 micro-hotels, 236 small hotels, 194 medium hotels, 180 large hotels, and 439 macro-hotels. Overall, 444 hotels were classified as MSMEs (Micro, Small, and Medium Enterprises), representing 42% of all registered hotels, while 619 hotels were categorized as large and macro enterprises, accounting for 58% of the total. Moreover, we identified 223 hotels classified as part of a chain of hotels and 840 as individual hotels. Despite most hotels in the Turismo de Portugal dataset had corresponding Tax Identification Numbers and in the Orbis dataset, not all financial variables were available across the period 2009–2015. This reduced the number of potential observations for study, a situation also seen in other relevant studies when using financial variables. For instance, Z-Score measure is based on three financial components and is only calculated when all components are available. For Z-Score measure, this reduced the number of observations to 1,930 across the period and similar level of observations are seen for other financial control variables. Nevertheless, the total number of observations used in this study in comparison to the total number of hotels during the same period is still high and comparable to other similar studies. In this sense, the sample used in this study is highly representative of the hotel sector allowing to generalize findings to the entire population with a reasonable degree of accuracy. A direct generalization seems appropriate for Southern European countries with similar tourism profiles such as Spain, Italy and Greece, for mature tourism markets with mixed leisure and cultural tourism, as well as countries with comparable economic development and credit markets. Portugal is a mature tourism market with diverse tourism types (beach, cultural, urban, wine), making it potentially representative of mixed tourism economies. Basic relationships between location, demand, revenue stability, and credit risk likely operate similarly across markets. Hotels everywhere need customers, and proximity to demand generators matters universally as fundamental economic principles. In addition, as tourism industry commonalities, standard hotel operational metrics (e.g. occupancy), financing structures, and credit risk assessment frameworks are relatively consistent globally, especially within developed markets.

Dependent variable

Z-Score measure is used to assess the hotels' credit risk (Vivel-Búa et al., 2018). The Z-score is chosen also because of its widespread use in empirical research (Boyd & Runkle, 1993; De Nicoló et al., 2004; Michalak & Uhde, 2012). A higher Z-Score implies a lower credit risk. The Z-Score formula incorporates three key components: profitability; capital structure; and profit volatility ($f_{roaa_volatility_w}$, σ). Profitability, measured by Return on Assets Average (ROAA), reflects how efficiently each hotel utilizes its assets to generate earnings, with higher values indicating better performance (f_{roaa_w} , μ). Capital structure, represented by the equity capital to total assets ratio, assesses the proportion of total assets financed by equity rather than debt, where a higher ratio indicates stronger financial health and a reduced reliance on external borrowing ($f_{equitycapital_total_assets_w}$, k). Finally, profit volatility (σ) captures fluctuations in ROAA over time, and for greater volatility higher risk is expected. Inconsistent profits can hinder an enterprise's ability to meet financial obligations, thereby increasing credit risk. By combining these elements, the Z-Score provides a comprehensive evaluation of credit risk, enabling the identification of hotels with elevated risks of financial distress.

$$Z - score = \frac{\mu + k}{\sigma} \quad (2)$$

μ = return on average assets (ROAA); k = capital ratio (capital level/total assets); σ = standard deviation (ROAA volatility).

The Z-score is used as the dependent variable in the regression analysis, using spatial distance from hotels to touristic points as the main variables of interest, as well as financial and hotel-specific characteristics as control variables. The hotel dataset is structured as a panel, combining cross-sectional and time-series dimensions by observing multiple hotel characteristics over a five-year period. Orbis dataset was used to collect the financial variables to calculate the Z-Score. We calculated both Z-Score and Z-Score Winsorized at a 5% significance level (f_{zscore_w}) as robustness-checks. While Winsorization mitigates the influence of extreme outliers and preserves the hotel sample, it may also lead to an underestimation of credit risk during periods of financial stress and may mask signals of true financial distress or exceptional performance among hotels.

Distance between each hotel's location and touristic location

Distance-based variables were constructed using Euclidean distances, following (Vivel-Búa et al., 2018). These variables are included as the main independent measures to evaluate the impact of location on hotels' creditworthiness. We developed the variable **location** (distance, in kilometers,

between hotels and tourist attractions and also per type of tourist attractions, namely, beaches, castles, and churches). We used the VisitPortugal government official website to identify all relevant tourist points across the country. This approach ensured that the geographic data collected was both systematic and reliable for further analysis. We developed the following five-distance-based variables. This method aligns with the geographic information system (GIS) approach described by [Sholeh et al. \(2017\)](#), where the authors demonstrate how the use of **latitude and longitude coordinates** enables the identification, storage, and spatial visualization of tourism-related locations through Google Maps API.

- Location 1 (ht_location_1) as the distance between each hotel and the nearest tourist attraction.
- Location 2 (ht_location_2) as the average distance between each hotel and the three nearest tourist attractions.
- Location 3 (ht_location_3) as the average distance between each hotel and the six nearest tourist attractions.
- Location 4 (ht_location_4): as the average distance between each hotel and the nine nearest tourist attractions.
- Location 5 (ht_location_5): as the average distance between each hotel and the twelve nearest tourist attractions.

Distances were calculated using the Euclidean distance ([Maria et al., 2020](#)).

$$D = \sqrt{(\phi_2 - \phi_1)^2 + (\lambda_2 - \lambda_1)^2} \quad (1)$$

Where, D = Euclidean distance in degrees, ϕ_i = latitude location i, λ_i = longitude location i., and the distance D in degrees is converted to kilometers by multiplying D by 111,319 km (1-degree earth = 69 miles $\sim$ 111,319 km).

The Euclidean is a heuristic function obtained based on direct distance free of obstacles. Previous research ([Vível-Búa et al., 2018](#)) used also Euclidean method to measure the distance from touristic centers (a specific location that includes and groups several touristic places) to their respective hotels. In our study we use the same distance formula, but with greater precision, by considering the specific distance from each touristic point (and not only touristic center's locations in general) to the hotels. We strengthening this location methodology and we further enhanced the analysis as mentioned before by also classifying tourist attractions into three distinct categories: beaches, castles, and churches. In this way, we created five additional locations for each of the three categories (i.e. ht_location_beach1 to ht_location_beach5; ht_location_castle1 to ht_location_castle5; ht_location_church1 to ht_location_church5, i.e. following the same methodology).

Control Variables

We identified and analyzed detailed information on the characteristics of the Portuguese hotels. The variables that were used in detail in this study providing the characteristics of the hotels include: star classification (ht_quality_stars, defined from 1 to 5), events capacity (ht_events_meetings, defined as the maximum number of seats in room events). Supportive variables (specifically room occupancy and bed occupancy rates) were used to construct additional control variables related to market share, market concentration; occupancy, and seasonality, following [Vível-Búa et al., 2018](#). Firstly, we created the market share variable by region NUTS II and NUTS III (f_mkt_share_nutsii; and f_mkt_share_nutsiii) defined as each hotel's share of total annual revenue within its respective region (f_tot_oper_rev_turn_nutsii; and f_tot_oper_rev_turn_nutsiii) from 2010 to 2015. Due to confidentiality constraints, hotel-level room availability and occupancy data were not accessible to calculate market share at room or occupancy level, so annual revenues at hotel and regional-level were used. We aggregated annual revenue data by region. Specifically, total revenues were calculated for each year and region. Secondly, market concentration (f_mkt_share_conc_nutsii; and f_mkt_share_conc_nutsiii) was computed by squaring each hotel's market share (f_mkt_share_pwr_nutsii; and f_mkt_share_pwr_nutsiii), summing these squared values by year and region. These values were then assigned to each hotel based on its corresponding year and region. Thirdly, the occupancy variable was developed using monthly room occupancy rates. For each month (January to December), data were collected at region level (ht_room_occup_rt_nutsii_1_to_12; and ht_room_occup_rt_nutsiii_1_to_12). These monthly values were used to calculate annual averages (ht_room_occup_nutsii; and ht_room_occup_nutsiii). These averages were linked to each hotel based on region and year. Finally, to measure seasonality (ht_room_occup_sea_nutsii; and ht_room_occup_sea_nutsiii) we used the variance of monthly occupancy rates by region and year. These values were then matched to the hotels according to their location and reporting year.

We utilized Orbis database to obtain comprehensive financial information about the Portuguese hotels included in our analysis from 2009-2015. The financial variables used in this study to characterize hotels include: Total assets (f_ln_total_assets_w), which represent the resources owned by the hotel (i.e., the sum of fixed and current assets), were transformed using the natural logarithm and Winsorized at a 5% significance level; the debt ratio (f_debt_r), calculated as the ratio of total liabilities to shareholders' equity; the efficiency ratio (f_efficiency_r), defined as the ratio of net income to total assets (i.e. the traditional Return on Assets or ROA); the growth ratio (f_growth_r), which measures the ratio of hotel revenues in the respective period relative to the previous period; the liquidity ratio (f_liquidity_r), computed as the ratio of current assets (excluding inventories) to current liabilities; and finally, the EBIT margin, i.e. the ratio of earnings before interest and taxes to operating revenues (f_profit_r), which captures operational profitability.

Descriptive analysis

Table 01 presents the summary statistics for the dependent variable Z-Score and control variables. The correlation analysis between (f_zscore_w) and financial control variables shows expected positive relationships regarding liquidity ratio, earnings before interest and taxes margin and efficiency ratio. Similarly, expected negative relationship is shown regarding debt ratio. Total assets ratio and growth ratio show a negative relationship; however, a non-linear relationship is often seen in literature regarding probabilities of default.

Table 01 – Descriptives – Dependent variable and control variables

Variables	N	Mean	Std. dev.	Min	Max
f_zscore_w	1,930	0.33	0.87	-2.29	20.90
f_debt_r	1,940	0.92	4.41	-56.56	124.97
f_efficiency_r	1,916	0	0.23	-3.44	5.58
f_growth_r	995	1,311.73	29,229.28	-1.21	796,338.10
f_liquidity_r	1,913	8.97	184.73	0	7,972.70
f_ln_total_assets_w	1,930	14.21	1.74	7.82	16.15
f_mkt_share_conc_nutsii	6,378	0.37	0.34	0	1
f_mkt_share_conc_nutsiii	6,378	0.35	0.34	0	1
f_mkt_share_nutsii	1,893	0.03	0.11	-0.02	1
f_mkt_share_nutsiii	1,893	0.07	0.17	-0.29	1
f_profit_r	1,828	2.60	21.38	-97.86	97.97
ht_events_meetings	7,441	107.17	229.00	0	3,000
ht_quality_stars	7,441	3.30	0.98	1	5
ht_room_ocup_nutsii	6,378	0.50	0.11	0.34	0.75
ht_room_ocup_nutsiii	6,378	0.49	0.13	0.22	0.75
ht_room_ocup_sea_nutsii	6,378	0.02	0.01	0.01	0.06
ht_room_ocup_sea_nutsiii	6,222	0.03	0.01	0.00	0.06

The analysis of correlation coefficients between the variable (f_zscore_w) and the location-based distance indicators exhibit, as expected, negative correlations. This shows that the lower the distance from the hotels' location from the touristic location, the highest the Z-Score (i.e. lower credit risk). Similar patterns are observed for distances from types of touristic points, namely beaches, castles and churches.

Table 02 – Descriptives – Location variables

Variables	Number of hotels	Mean	Std. dev.	Min	Max
ht_location_1	882	3.44	5.78	0.04	44.40
ht_location_2	882	5.65	7.83	0.08	53.64
ht_location_3	882	8.26	9.48	0.24	58.58
ht_location_4	882	10.93	10.90	0.55	67.78
ht_location_5	882	32.42	73.37	0.79	308.96
ht_location_beach1	411	14.02	14.75	0.05	73.83
ht_location_beach2	411	17.87	16.81	0.24	83.50
ht_location_beach3	411	22.37	20.33	0.35	96.53
ht_location_beach4	411	34.47	37.63	0.64	162.22
ht_location_beach5	411	53.18	87.90	1.26	368.43
ht_location_castle1	96	87.51	259.70	0.11	1,037.65
ht_location_castle2	96	103.71	259.92	7.47	1,044.10
ht_location_castle3	96	144.69	311.56	15.82	1,177.70
ht_location_castle4	96	167.39	341.08	25.98	1,512.19

ht_location_castle5	96	183.38	357.41	31.78	1,684.60
ht_location_church1	375	7.02	10.85	0.04	80.34
ht_location_church2	375	56.50	171.59	0.08	680.93
ht_location_church3	375	73.05	215.26	0.26	854.76
ht_location_church4	375	81.57	230.43	0.61	922.07
ht_location_church5	375	87.84	238.20	0.85	958.92

Dynamic panel data

Previous studies have employed pooled Ordinary Least Squares (OLS), probit, and logit models to examine the determinants of credit risk. We are using a continuous credit risk measurement, namely Z-Score (i.e. not binary and not suitable for Probit or Logit models). In addition, hotels have unobserved characteristics that affect credit risk and some may not be captured by our variables, for instance: management quality and experience; brand reputation and customer loyalty; organizational culture and risk management practices. Pooled OLS assumes that those potential unobserved factors are uncorrelated with our explanatory variables and this assumption would almost be certainly violated. In this study, the one-step System Generalized Method of Moments (System GMM) estimator is applied to address potential endogeneity, as suggested by [Lado-Sestayo, Otero-Gonzalez, Vivel-Búa, and Martorell-Cunill \(2016\)](#), to avoid biased results and enhance robustness. This estimator is particularly suitable for dynamic panel data where the Z-score depends on its lagged values. The Arellano–Bond test for autocorrelation is conducted to validate the specification, ensuring that residuals do not exhibit second-order serial correlation and that the moment conditions underlying the estimator are satisfied. This approach captures unobserved heterogeneity and dynamic effects in the relationship between explanatory variables and credit risk, thereby strengthening the reliability of the analysis. In this dynamic panel methodology, lags of the dependent variable as additional explanatory variables are used ($Creditrisk_{it(t-1)}$), capturing the persistence of credit risk, and controls for endogeneity are implemented as instrumental variables. Firm-specific effect (η_i) and time-specific effect (dt) were also included to control for, respectively, unobservable heterogeneity and the effect of macroeconomic variables. The dynamic panel is given by:

$$Creditrisk_{it} = \alpha + \rho Creditrisk_{it(t-1)} + \beta_k X_{k,it} + \eta_i + dt + \varepsilon_{it} \quad (4)$$

where i = hotel; t = year; α is the constant term; $Creditrisk_{it(t-1)}$ = lagged dependent variable; $X_{k,it}$ = vector of k explanatory variables (location, firm-specific, industry-specific, and region determinants); η_i = unobserved firm-specific (i) fixed effects effects; dt = time-specific (t) fixed effects; ε_{it} = idiosyncratic error term; ρ and β_k are parameters to be estimated.

The one-step System GMM provides important advantages in this study, namely: it allows to control for unobserved heterogeneity; addresses endogeneity of explanatory variables; handles the dynamic nature of credit risk; is suitable for panels with few years and a large number of firms; and it is more efficient than difference GMM when variables are persistent such as the case for credit risk. Typically, credit risk variables are quite persistent because hotels with poor credit quality tend to remain risky over time, financial distress is a gradual process, not sudden, and structural factors such as location of the hotels and management quality persist across years.

4. Results

This study provides empirical evidence on the relationship between individual hotel credit risk, measured by the Z-score measure, and distances from hotel location to touristic places. The findings indicate that, when considering several different types of touristic places, the lower the distance between hotels and those nearest tourist attractions, the higher is the Z-Score (i.e. reduction of credit risk). [Table 03](#) presents the determinants of credit risk in the Portuguese hotel sector by different definitions of touristic attractions. When considering the hotels' distance to only one touristic place (column 1) or to the average of the three and six nearest touristic places (columns 2 and 3), the relationship between the distance from hotels to touristic places and Z-Score is not significant. However, when considering the average distance to the twelve nearest tourist attractions ($ht_location_5$, in column 5), the coefficient (coef. = -0.010 , p -value < 0.01) is significant. showing that the lower the distance between hotels and those nearest twelve tourist attractions, the higher is the Z-Score (i.e. reduction of credit risk). These results are in line with the findings of [Vivel-Búa et al. \(2018; coef. = \$-0.019\$, \$p\$ -value \$< 0.01\$ \)](#). A one-kilometer reduction in this average distance is associated with an increase of 0.01 in the hotel's Z-score, indicating lower credit risk. In addition, the results show that there are threshold effects confirming a necessary critical mass for hotel financial performance and viability. Below a certain threshold of major tourism attractions, location may not be viable for tourism-dependent hotels, increasing credit risk. Whereas above a certain threshold of tourist places, the geographic area becomes a recognized tourist destination, and closest hotels improve the credit quality indicators.

Critical mass refers to the minimum concentration of tourist attractions necessary for a location to become a viable, self-sustaining tourist destination

that can support profitable hotel operations with acceptable credit risk levels. The concept draws from agglomeration theory in economic geography where individual attractions in isolation generate limited, sporadic visits insufficient to justify overnight stays and cannot support dedicated tourist infrastructure. This concept of critical mass is important to take into account from a credit risk perspective. Below a certain threshold, hotels face insufficient demand to maintain viable occupancy rates, high revenue volatility and seasonality and elevated probability of financial distress and significant credit risk. Above a certain threshold, in case of clustered attractions, a critical mass of activities exists and justify multi-day visits and overnight stays, attracting complementary businesses such as restaurants, shops, and transport, and generating positive externalities and spillover effects.

In the context of critical mass thresholds and location viability, it is expected that total assets show a negative relationship with credit quality, denoting significance at the 1% level. Hotels with larger total assets in locations below critical mass face a particularly dangerous mismatch, namely higher fixed costs and insufficient demand produce a massive operating leverage and a higher probability of distress and lower credit scoring. In the same vein, higher EBIT margin (f_profit_r) from cost-cutting together with revenue decline due to hotel's location with closer touristic attractions below a certain threshold may show an unsustainable operational model and deteriorating service quality, showing a lower credit quality score, denoting significance at the 1% level. Hotels in below-threshold locations facing revenue pressure may engage in defensive cost-cutting and are often forced into volume and discount strategy. Given insufficient touristic attractions, these hotels may chase occupancy through discounts, cut costs to maintain margins via service deterioration and this may mask weak competitive positions. For retail sector, Altman & Sabato (2008) shows that margin expansion during revenue contraction predicts bankruptcy, whereas for the hotel sector Gu & Kim (2002) shows that margin volatility (not just level) predicts hotel financial distress. We show that margin expansion may predict credit risk conditional on hotels' location viability. In addition and theoretically coherent when paired with the EBIT margin finding, we can find also a negative relationship between liquidity ratio and credit quality, denoting significance at the 5% level, and signaling together a powerful distress signal combination. Hotels facing financial stress simultaneously manage two key ratios that stakeholders monitor, EBIT margin for operational appearance and liquidity ratio for financial appearance in a defensive cash hoarding from cost savings. This builds apparent financial cushion and signals to creditors that the hotel can meet obligations, however the cash hoarding may be provoking underinvestment, competitive erosion and a future revenue decline, therefore reducing credit scoring despite "good" ratios. In crisis-periods, precautionary cash holdings in uncertainty (Opler et al., 1999) and defensive behaviors (Campello et al., 2010) are more evident, with hotels engaging in aggressive cost-cutting while hoarding liquidity (Okumus and Karamustafa, 2005). The efficiency ratio, defined as the ratio of net income to total assets (i.e. the traditional Return on Assets or ROA), as expected shows a positive sign, denoting significance at the 1% level. High efficiency ratio relates to better credit scoring (lower credit risk) and this is consistent and complements the previous findings from financial controls. High efficiency ratio combines profitability with appropriate scale and adjusts for asset base, controlling the potential overleveraging problem (large assets negatively related to credit quality) mentioned before and reveals true productivity of capital employed. That is, assets are inherently bad, only the unproductive assets are bad and efficiency ratio captures this distinction and genuine hotels' performance from distress. Therefore, efficiency ratio positively correlates with good credit scoring, even though absolute asset size negatively correlates with the same good credit scoring. The advantage of the efficiency ratio, as a combined metric, is that requires both profitability and productivity (and cannot be gamed by just one dimension) functioning as a holistic measure of business health. Consistent with Altman's (1968) seminal work on bankruptcy prediction and previous results in the hotel sector (Vível-Búa et al., 2018), we find efficiency ratio (i.e. return on assets) significantly and positively predicts hotel credit quality (i.e. lower credit risk).

The rate of occupancy at regional level, as expected, shows that high regional occupancy relates to better credit scoring (lower credit risk), denoting significance at the 1% level. In case of high regional occupancy, this reflects a strong underlying tourism demand that all hotels benefit provoking a better revenue generation and improving credit scores. Moreover, a high occupancy at regional level enables premium pricing with hotels having pricing power to raise average daily rates without losing bookings. High occupancy at regional level also reduces cash flow risk and revenue's volatility. However, this positive relationship is highly heterogeneous and depends critically on hotel-specific characteristics (Chung, W., & Kalnins, A., 2001; Shoval et al., 2011). For hotels in above-threshold touristic locations, with moderate asset bases, regional occupancy improvements translate into meaningful credit risk reductions. Conversely, for large hotels in below-threshold locations, regional market strength provides minimal protection against elevated credit risk. Therefore, regional occupancy is useful but may be an insufficient predictor of individual hotel credit risk without considering location, quality service, size, and financial management behaviors. Lenders cannot rely on regional market strength alone and should evaluate hotel-specific factors to accurately assess credit risk.

The quality stars of hotel-specific factors, as expected, show that higher number of stars relates to better credit scoring, denoting significance at the 5% level. This is consistent with the previous findings. Star rating shows the quality positioning and strategic choice of each hotel, which differentiates from financial engineering, EBIT margin management, liquidity hoarding and possible overleveraged large assets. Start rating may reflect a genuine sustainable competitive advantage that cannot be artificially inflated. For instance, 5-star hotels use premium average daily rates, driving revenues even at lower occupancy and achieving high revenues per available room (Israeli, A. A., 2002; Assaf et al., 2012). These 5-star hotels are less vulnerable to demand shocks, with wealthy clientele less elastic, allowing a stronger debt service capacity and higher credit scoring (lower credit risk). The models

robustness are supported by strong Wald χ^2 statistics (e.g. column 5, 804.65), significant first-order serial correlations (e.g. column 5, $AR(1) = -2.50$, $p\text{-value} < 0.01$), and no evidence of second-order autocorrelations (e.g. column 5, $AR(2) = 0.48$, $p\text{-value} > 0.1$).

Table 03: Determinants of credit risk in the Portuguese hotel sector by different definitions of location (distances from hotels to the nearest tourist attractions)

Variables	Dynamic Panel Data				
	Dependent Variable:				
	Z-score				
	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)
	(1)	(2)	(3)	(4)	(5)
L.Z-score_w	1.584 (0.115) ***	1.588 (0.113) ***	1.586 (0.111) ***	1.504 (0.132) ***	1.457 (0.117) ***
ht_location_1	0.027 (0.045)	-	-	-	-
ht_location_2	-	0.031 (0.045)	-	-	-
ht_location_3	-	-	0.032 (0.046)	-	-
ht_location_4	-	-	-	-0.047 (0.038)	-
ht_location_5	-	-	-	-	-0.010 (0.002) ***
f_mkt_share_conc_nutsii	0.149 (0.887)	0.092 (0.878)	0.030 (0.884)	0.357 (1.079)	0.134 (0.959)
ht_room_ocup_sea_nutsii	28.273 (28.280)	34.397 (31.233)	38.907 (34.523)	2.256 (36.451)	-19.589 (31.453)
ht_events_meetings	0.001 (0.001) **	0.001 (0.001) **	0.001 (0.001) **	0.001 (0.001) *	0.001 (0.001) **
f_ln_total_assets_w	-0.293 (0.170) *	-0.298 (0.162) *	-0.299 (0.160) *	-0.397 (0.191) **	-0.464 (0.170) ***
f_liquidity_r	-0.011 (0.005) **	-0.012 (0.005) **	-0.012 (0.006) **	-0.004 (0.005)	-0.009 (0.003) **
f_debt_r	0.056 (0.072)	0.053 (0.071)	0.057 (0.070)	0.045 (0.086)	0.036 (0.077)
f_efficiency_r	1.838 (0.428) ***	1.785 (0.450) ***	1.747 (0.478) ***	2.391 (0.533) ***	2.029 (0.378) ***
f_profit_r	-0.048 (0.014) ***	-0.047 (0.014) ***	-0.046 (0.014) ***	-0.066 (0.017) ***	-0.057 (0.012) ***
f_growth_r	-0.001 (0.001) *	-0.001 (0.001) *	-0.001 (0.001)	-0.001 (0.001) **	-0.001 (0.001) ***
ht_room_ocup_nutsii	5.170 (2.820) *	5.375 (2.849) *	5.355 (2.815) *	4.374 (3.239)	11.442 (3.468) ***
ht_quality_stars	0.171 (0.168)	0.153 (0.174)	0.146 (0.178)	0.321 (0.194) *	0.341 (0.162) **
f_mkt_share_nutsii	-0.005 (0.947)	0.026 (0.919)	0.081 (0.904)	0.579 (1.156)	1.760 (1.089)
2011.year	-1.104 (0.662) *	-1.072 (0.660)	-1.031 (0.671)	-1.436 (0.807) *	-1.261 (0.703) *
2012.year	-0.315 (0.419)	-0.251 (0.454)	-0.218 (0.478)	-0.845 (0.508) *	-0.669 (0.368) *
2013.year	0.081 (0.431)	0.152 (0.462)	0.200 (0.493)	-0.340 (0.556)	0.133 (0.440)
2014.year	-0.041 (0.162)	-0.027 (0.165)	-0.021 (0.168)	-0.154 (0.184)	0.183 (0.174)
Constant	-0.032 (2.507)	-0.272 (2.569)	-0.428 (2.675)	2.758 (2.708)	0.071 (2.062)
Observations (no. of hotels)	960	960	960	960	960
Wald (χ^2)	926.590	948.820	958.580	645.220	804.650
AR(1)	-3.030 ***	-3.110 ***	-3.130 ***	-2.380 **	-2.540 ***
AR(2)	0.660	0.690	0.700	0.190	0.480
No. instruments	41	41	41	41	41

Notes: Coef. denotes the coefficient of explanatory variables and Std. error represents the heteroscedasticity-robust asymptotic standard error. AR(1) is a serial correlation test of order 1 (and AR(2) of order 2 using residuals in first differences, under the null hypothesis of no serial correlation. Wald (χ^2) is a test of the joint significance of the reported coefficients, under the null hypothesis of no relationship. ***, **, * denote significance at the 1%, 5% and 10% levels, respectively.

After providing empirical evidence on the relationship between individual hotel credit risk and distances from hotel location to touristic places in general, this study provides more details regarding the type of touristic attractions. We study three separate types of touristic attractions, namely: beaches, castles, and churches. The findings indicate that the relationship between individual hotel credit risk and distances from hotel location to touristic places is not linear and sometimes even inverted. Distances to different types of tourist attractions proxy for market segment exposure — leisure beach, cultural tourism, and religious tourism—each showing different shock sensitivities.

The findings for market segment exposure leisure beach indicate that the proximity of the hotels to beaches has a significant impact on the level of credit

risk, however the relationship is not linear and, in fact, inverted depending on the number of beaches. [Table 04](#) shows the determinants of credit risk in the Portuguese hotel sector by different definitions of beaches' location (distances from hotels to the nearest beaches). This reveals that distance to beaches affects credit risk differently depending on whether you consider distance to the nearest few beaches (columns 1 and 2, showing for higher distances a positive relationship with good credit scoring) versus many beaches (columns 4 and 5, showing for higher distances a negative relationship with good credit scoring). This would reveal a critical proximity-density trade-off in coastal hotel location strategy. Whereas the distance to the nearest beach and three nearest beaches measures immediate coastal crowding and competition (ht_location_1 and ht_location_2), the distance to the nine nearest beaches and twelve nearest beaches (ht_location_4 and ht_location_5) measures the regional beach density and coastal tourism ecosystem. Consistent with [Rigall-I-Torrent and Fluvía's \(2011\)](#) hedonic pricing findings, we document a similar non-monotonic pattern for credit risk, where moderate distances (i.e. not too close) from immediate beaches minimize financial distress while maintaining beach accessibility (i.e. not too far). In addition, extending [Canina et al. \(2005\)](#) agglomeration framework to coastal locations, we find an analogous inverted U-shaped relationship: moderate distances from clustered beach hotels optimize credit risk by balancing beach accessibility against direct competitive pressure. The control variables show the same relationships described before. The models robustness are supported by strong Wald χ^2 statistics (e.g. column 5, 734.52), significant first-order serial correlations (e.g. column 5, AR(1) = -2.50, p-value < 0.01), and no evidence of second-order autocorrelations (e.g. column 5, AR(2) = -0.26, p-value > 0.1).

Table 04: Determinants of credit risk in the Portuguese hotel sector by different definitions of beaches' location (distances from hotels to the nearest beaches)

Variables	Dynamic Panel Data				
	Dependent Variable:				
	Z-score				
	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)
L.Z-score_w	1.603 (0.099) ***	1.591 (0.100) ***	1.569 (0.104) ***	1.472 (0.133) ***	1.462 (0.122) ***
ht_location_beach1	0.040 (0.017) **	-	-	-	-
ht_location_beach2	-	0.032 (0.015) **	-	-	-
ht_location_beach3	-	-	0.013 (0.013)	-	-
ht_location_beach4	-	-	-	-0.018 (0.006) ***	-
ht_location_beach5	-	-	-	-	-0.008 (0.002) ***
f_mkt_share_conc_nutsii	0.475 (0.839)	0.329 (0.840)	0.167 (0.876)	0.099 (1.094)	0.105 (1.004)
ht_room_ocup_sea_nutsii	64.406 (30.844) **	40.714 (26.732)	23.469 (26.588)	9.631 (33.556)	-2.341 (31.372)
ht_events_meetings	0.001 (0.001) **	0.001 (0.001) ***	0.001 (0.001) **	0.001 (0.001) **	0.001 (0.001) **
f_ln_total_assets_w	-0.315 (0.143) **	-0.322 (0.145) **	-0.326 (0.152) **	-0.446 (0.194) **	-0.458 (0.178) **
f_liquidity_r	-0.013 (0.003) ***	-0.011 (0.003) ***	-0.010 (0.003) ***	-0.007 (0.004) *	-0.009 (0.004) **
f_debt_r	0.115 (0.070)	0.136 (0.076) *	0.110 (0.086)	-0.049 (0.096)	-0.010 (0.083)
f_efficiency_r	1.447 (0.400) ***	1.454 (0.417) ***	1.715 (0.441) ***	2.364 (0.452) ***	2.134 (0.398) ***
f_profit_r	-0.035 (0.013) ***	-0.036 (0.013) ***	-0.044 (0.014) ***	-0.070 (0.015) ***	-0.062 (0.013) ***
f_growth_r	-0.001 (0.001) *	-0.001 (0.001) *	-0.001 (0.001) **	-0.001 (0.001) ***	-0.001 (0.001) ***
ht_room_ocup_nutsii	5.172 (2.516) **	5.918 (2.605) **	5.426 (2.768) *	8.687 (3.626) **	10.820 (3.538) ***
ht_quality_stars	0.160 (0.139)	0.177 (0.140)	0.204 (0.146)	0.299 (0.183)	0.311 (0.168) *
f_mkt_share_nutsii	0.006 (0.854)	-0.134 (0.872)	-0.112 (0.939)	1.930 (1.304)	1.997 (1.171) *
2011.year	-1.164 (0.605) *	-1.127 (0.612) *	-1.107 (0.646) *	-1.312 (0.803)	-1.258 (0.736) *
2012.year	-0.141 (0.343)	-0.208 (0.340)	-0.340 (0.355)	-0.693 (0.423)	-0.628 (0.383)
2013.year	0.000 (0.377)	0.085 (0.384)	0.077 (0.409)	0.015 (0.500)	0.134 (0.461)
2014.year	0.026 (0.144)	0.058 (0.153)	-0.002 (0.166)	0.008 (0.183)	0.135 (0.177)
Constant	-1.264 (2.007)	-1.023 (2.021)	0.024 (2.090)	1.169 (2.337)	0.161 (2.156)
Observations (no. of hotels)	960	960	960	960	960
Wald (χ^2)	1074.060	1050.240	950.520	616.740	734.520
AR(1)	-3.010 ***	-2.920 ***	-2.890 ***	-2.370 **	-2.500 **
AR(2)	0.820	0.870	0.760	-0.010	-0.260
No. instruments	41	41	41	41	41

Notes: Coef. denotes the coefficient of explanatory variables and Std. error represents the heteroscedasticity-robust asymptotic standard error. AR(1) is a serial correlation test of order 1 (and AR(2) of order 2 using residuals in first differences, under the null hypothesis of no serial correlation. Wald (χ^2) is a test of the joint significance of the reported coefficients, under the null hypothesis of no relationship. ***, **, * denote significance at the 1%, 5% and 10% levels, respectively.

The findings for market segments exposure cultural tourism and religious tourism indicate, as expected, that the proximity of the hotels to castles and churches has a significant impact on the level of credit risk, the relationship is linear and not dependent on the number of those touristic attractions. Hotels located closer to both castles and churches tend to exhibit better quality scoring (i.e. the lower the distance between hotels and castles and churches, the higher the Z-Scores and lower the credit risk). The control variables show the same relationships described before. The models robustness are supported by strong Wald χ^2 statistics, significant first-order serial correlations (AR(1), p-value < 0.01), and no evidence of second-order autocorrelations (AR(2), p-value > 0.1).

In order to reveal more complex relationships on different definitions of beaches' location (distances from hotels to the nearest beaches) that may be hidden when variables are analyzed in isolation, some important interactions were developed, allowing for a more nuanced understanding of outcomes and precise predictions. Table 05, the interactions between different definitions of beaches' location (i.e. ht_location_beach1 to ht_location_beach5) and hotels' EBIT margins (f_profit_r) are presented. Previous studies showed how location affects interpretation of financial metrics. Dalbor and Upneja, (2004) showed that high profitability margins signal firms' health in stable markets, however same margins may signal firms' distress in competitive markets and location-type may moderate ratio interpretation. Prior research demonstrates that location context moderates the interpretation of financial metrics in hospitality. Canina et al. (2005) provide direct evidence that profitability margins exhibit location-contingent effects: in high-competition locations, elevated margins predict declining performance, consistent with distress-driven cost-cutting, while in low-competition locations, high margins predict sustained success, indicating genuine efficiency. This paradoxical pattern is validated by Enz and Canina's (2005) natural experiment following 9/11, where urban competitive hotels showed rising margins alongside deteriorating performance, while suburban properties achieved sustainable margin improvements. Chen and Chang (2012) further document two distinct high-margin clusters—distressed properties in competitive locations versus efficient operators in stable markets—with location emerging as the critical moderator. Industry practitioners confirm this context-dependence, with 94% of hotel CFOs adjusting margin benchmarks by property type and location (Singh & Schmidgall, 2002), viewing high margins as concerning in resort contexts but desirable in suburban settings.

Consistent with prior research documenting location-contingent margin interpretation (Canina et al., 2005; Enz & Canina, 2002; Chen & Chang, 2012), we find that at close proximity between hotels and beaches, high profit margins increase credit risk (for each 1% increase in profit margin, the Z-score reduces 0.07, p < 0.01). The significant and positive signs of the interactions between different distances (from the hotels to the beaches) and EBIT margin show that the negative relationship between EBIT margin and good credit quality (i.e. positive relationship with credit risk, as a distress signal), becomes weaker (less negative) as distance from beaches increases. The finding aligns with financial distress literature (Asquith et al., 1994; Canina et al., 2005), suggesting that elevated margins signal distress in competitive locations but efficiency in stable markets. In practical terms, on the one hand, hotels close to beaches that show high EBIT margins may be considered of very high credit risk (distress signal amplified) and, on the other hand, hotels far from beaches (not beach-focused) that show high EBIT margins may be considered of lower risk (legitimate efficiency, e.g. due to stable operations and not distress). The positive interactions capture this distinction in credit risk and provide novel evidence of context-dependent financial ratio interpretation. The specification now correctly distinguishes distress from efficiency, and the interaction term is essential for correctly interpreting EBIT margins. The models robustness are supported by strong Wald χ^2 statistics (e.g. column 1, 734.52), significant first-order serial correlations (e.g. column 1, AR(1) = -2.50, p-value < 0.01), and no evidence of second-order autocorrelations (e.g. column 1, AR(2) = -0.26, p-value > 0.1).

Table 05: Interactions between hotels' location (distances from hotels to the nearest beaches) and EBIT margin (f_profit_r)

Variables	Dynamic Panel Data				
	Dependent Variable:				
	Z-score				
	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)
	(1)	(2)	(3)	(4)	(5)
L.Z-score_w	1.466 (0.134) ***	1.442 (0.119) ***	1.417 (0.117) ***	1.436 (0.150) ***	1.446 (0.132) ***
ht_location_beach1	0.083 (0.025) ***	-	-		
ht_location_beach1*f_profit_r	0.003 (0.001) ***	-	-		
ht_location_beach2	-	0.072 (0.020) ***	-		
ht_location_beach2*f_profit_r	-	0.002 (0.001) ***	-		
ht_location_beach3	-	-	0.059 (0.017) ***		
ht_location_beach3*f_profit_r	-	-	0.001 (0.001) ***		
ht_location_beach4	-	-	-	-0.014 (0.007) *	

ht_location_beach4*f_profit_r	-	-	-	0.001 (0.001)	
ht_location_beach5	-	-	-		-0.008 (0.002) ***
ht_location_beach5*f_profit_r	-	-	-		0.001 (0.001)
f_mkt_share_conc_nutsii	1.583 (1.122)	1.101 (0.978)	0.694 (0.954)	0.279 (1.222)	0.218 (1.088)
ht_room_ocup_sea_nutsii	59.845 (39.701)	51.736 (30.704) *	45.219 (29.112)	46.217 (45.052)	23.360 (40.760)
ht_events_meetings	0.001 (0.001) **	0.001 (0.001) ***	0.001 (0.001) ***	0.001 (0.001) *	0.001 (0.001) **
f_ln_total_assets_w	-0.453 (0.188) **	-0.435 (0.168) **	-0.414 (0.165) **	-0.480 (0.217) **	-0.494 (0.194) **
f_liquidity_r	-0.017 (0.005) ***	-0.014 (0.004) ***	-0.015 (0.004) ***	-0.010 (0.005) *	-0.010 (0.004) **
f_debt_r	0.045 (0.092)	0.031 (0.090)	0.017 (0.095)	-0.142 (0.125)	-0.069 (0.103)
f_efficiency_r	1.089 (0.525) **	0.781 (0.502)	0.676 (0.527)	2.280 (0.506) ***	2.158 (0.430) ***
f_profit_r	-0.067 (0.019) ***	-0.064 (0.017) ***	-0.067 (0.016) ***	-0.089 (0.021) ***	-0.074 (0.017) ***
f_growth_r	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001) **	-0.001 (0.001) **
ht_room_ocup_nutsii	4.702 (3.239)	7.076 (2.994) **	8.196 (3.051) ***	6.553 (4.291)	9.437 (4.009) **
ht_quality_stars	0.333 (0.185) *	0.234 (0.160)	0.171 (0.158)	0.283 (0.204)	0.319 (0.182) *
f_mkt_share_nutsii	0.463 (1.107)	-0.100 (0.998)	-0.829 (1.027)	1.320 (1.509)	1.784 (1.277)
2011.year	-2.146 (0.825) ***	-1.523 (0.707) **	-1.081 (0.698)	-1.397 (0.895)	-1.364 (0.800) *
2012.year	-0.912 (0.490) *	-0.526 (0.396)	-0.204 (0.385)	-0.578 (0.477)	-0.550 (0.420)
2013.year	-0.658 (0.519)	-0.192 (0.444)	0.108 (0.442)	-0.149 (0.567)	0.006 (0.510)
2014.year	-0.087 (0.188)	0.049 (0.176)	0.083 (0.180)	-0.186 (0.244)	-0.002 (0.227)
Constant	0.039 (2.607)	-1.115 (2.313)	-1.632 (2.286)	1.947 (2.652)	0.819 (2.398)
Observations (no. of hotels)	960	960	960	960	960
Wald (χ^2)	661.910	820.300	835.500	521.290	631.830
AR(1)	-2.690 ***	-3.070 ***	-3.310 ***	-2.130 **	-2.370 **
AR(2)	0.570	0.970	1.350	-0.080	0.160
No. instruments	41	41	41	41	41

Notes: Coef. denotes the coefficient of explanatory variables and Std. error represents the heteroscedasticity-robust asymptotic standard error. AR(1) is a serial correlation test of order 1 (and AR(2) of order 2 using residuals in first differences, under the null hypothesis of no serial correlation. Wald (χ^2) is a test of the joint significance of the reported coefficients, under the null hypothesis of no relationship. ***, **, * denote significance at the 1%, 5% and 10% levels, respectively.

Traditional credit models assume that high profitability margins are in principle good, however the same financial metric can signal opposite states depending on operational context. Our findings show that, in the hotel sector, high profitability margins are context-dependent and in case of high-stress contexts (such as hotels' beachfront competition) can signal credit risk instead. Profit margin interpretation flips across hotels-beaches distance spectrum and this enhanced margin interpretation should be taken into account by credit institutions (enhanced underwriting for loan applications; risk-adjusted pricing considering EBIT margin risk premium, as basis points added to base rate). In this sense, credit models for hotels' exposures should incorporate conditional interpretation of EBIT margin. This approach is also useful in the context of hotel operators strategic positioning in case of beachfront hotels needing to communicate to lenders the trade-off between reasonable high profit margins (not signaling distress and cross-cutting damaging guest experience) and consistent service investment (maintenance of service quality and reinvestments in property).

5. Conclusions

This paper offers an in-depth investigation into the determinants of credit risk using Z-Score within the Portuguese hotel industry using a representative sample of hotels in the period 2010–2015. It emphasizes the importance of integrating spatial factors in addition to financial-operational factors to accurately assess hotels' credit risk. The research highlights that hotels' location plays a significant role in influencing credit risk, with hotels situated closer to major tourist points generally exhibiting lower risk profiles. These results are in line with the findings of [Vivel-Búa et al. \(2018\)](#). In addition, by considering in our study different definitions of geospatial average locations (based on different number of touristic points), the results also show that there are threshold effects, confirming a necessary critical mass for hotel financial performance and viability. This concept of critical mass that we show in this study is important to take into account from a credit risk perspective. Below a certain threshold of major tourism attractions, location may not be viable for tourism-dependent hotels, increasing credit risk. Whereas above a certain threshold of tourist places, the geographic area becomes a recognized tourist destination, and closest hotels improve the credit quality indicators. Moreover, in the context of critical mass thresholds and location viability, we show that total assets have a negative relationship with credit quality (the higher the total assets, the lower the Z-Score). In the same vein,

hotels facing financial stress simultaneously manage two key ratios that stakeholders monitor, EBIT margin for operational appearance and liquidity ratio for financial appearance in a defensive cash hoarding from cost savings. This apparent financial cushion signals to creditors that the hotel can meet obligations, however the cash hoarding may be provoking underinvestment, competitive erosion and a future revenue decline, therefore reducing Z-Score (credit quality) despite showing good ratios from a traditional viewpoint. In the period of the study, between 2010 and 2015, the 2008 Global Financial Crisis and the impact of the 2010-2012 Euro-Area Sovereign Crisis, collectively weakened demand in the tourism and hospitality industries, increasing financial vulnerability for hotels, and reinforced this concept of critical mass.

We also studied that distances to different types of tourist attractions proxy for market segment exposure — leisure beach, cultural tourism, and religious tourism— each showing different shock sensitivities. The findings for market segments exposure cultural tourism and religious tourism indicate, as expected, that the proximity of the hotels to castles and churches has a significant impact on the level of credit risk, the relationship is linear and not dependent on the number of those touristic attractions. The findings for market segment exposure leisure beach indicate that the proximity of the hotels to beaches has a significant impact on the level of credit risk, revealing a critical proximity-density trade-off in coastal hotel location strategy. We document a non-monotonic pattern for credit risk, where moderate distances (i.e. not too close) from immediate beaches minimize financial distress while maintaining beach accessibility (i.e. not too far). In addition, extending Canina et al. (2005) agglomeration framework to coastal locations, we find an analogous inverted U-shaped relationship showing that moderate distances from clustered beach hotels optimize credit risk by balancing beach accessibility against direct competitive pressure.

In order to reveal more complex relationships on different definitions of beaches' location (distances from hotels to the nearest beaches) that may be hidden when variables are analyzed in isolation, some important interactions were developed. High EBIT margins generally signal credit risk (distress cost-cutting), however this relationship weakens for hotels far from beaches. Close to beaches, high margins indicate desperate cost-cutting in competitive markets (distress). Far from beaches, high margins indicate efficient lean operations in stable markets (health). For beaches at close proximity from hotels, high profit margins signals credit risk, consistent with distress cost-cutting in competitive beachfront markets. As distance increases between hotels and beaches locations, this positive relationship (high profit margins and credit risk) weakens where hotels with high margins instead signal operational efficiency and reduce credit risk instead. This crossover pattern demonstrates context-dependent interpretation. For identical margin levels signal opposite states (distress vs. efficiency) depending on hotels' competitive environment.

This study contributes to hotel financial distress literature by demonstrating that distress signals are location-contingent. It contributes also to operational efficiency literature by showing that the context determines what efficient means by quantifying context-dependent signal interpretation (differentiating efficient margin levels in a service investment context from efficient margin levels in a lean operations context). Business models are contingent on location and different optimal margin structures applies meaning that it may not be correct to benchmark efficiency across different competitive contexts. Moreover, it contributes to cost management literature by showing that cost-cutting interpretation depends on required service level. In case of high-service contexts, such as beachfront hotels, cost-cutting reduces margins and may signal financial risk, whereas in case of low-service contexts (for instance, more suburban hotels), cost optimization raises margins and may sign financial health. Therefore, margin increases can signal opposite states, and context needs to be considered. As policy and regulatory implications, this study contributes to enhance credit risk guidance by proposing that profitability margin interpretation must consider operational context and that location-adjusted interpretation should be used in risk ratings. The high profitability margins context-dependent are also an important concept for investors and respective due diligence enhancement and valuation adjustment.

6. References

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